The Projection Law: Model-Ensemble Errors Point at Their Binding Constraint

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Abstract

Purpose: Multi-model agreement is routinely used as confidence in materials simulation and beyond, yet models in a family share construction and training lineage. This work formalizes and tests the projection law: a model family acts as a projection operator, every well-fitted member shares one residual — a single direction in observable space — and that direction fingerprints whatever constraint binds the family. **Methods:** The theoretical core (normal-cone consensus theorem; participation-ratio gauge $\mathrm{PR}(d, \rho) = (\rho + d)^2/((\rho + 1)^2 + (d - 1))$ with explicit ribbon-collapse rate; ribbon/consensus decoupling) is machine-checked in Lean 4 with zero unproven obligations. Two factorial experiments were pre-registered with refutation conditions committed before execution: a 4×2 architecture-by-functional crossing of MatPES-trained foundation interatomic potentials evaluated on elastic constants of 15 cubic metals, and an analysis of 12 DFT implementations on 384 equation-of-state benchmarks from the ACWF verification effort. A two-tier replication kit verifies every statistic from committed raw data (NumPy only, seconds) and re-derives the raw data from public checkpoints (bit-exact). **Results:** Foundation-model errors cluster by training functional, not architecture (S_{func}=+0.317 vs. S_{arch}=−0.093, exact permutation p=0.029); r^{2}SCAN training rotates noble-metal C_{44} errors toward experiment as predicted. DFT-implementation errors cluster by pseudopotential table, not simulation code (S_{table}=+0.526 vs. S_{code}=+0.265, p=0.017); four independent plane-wave codes sharing one table align at 0.70–0.87. Neither registered refutation condition was triggered. Combined with the 559-potential classical baseline (within-family error correlation 0.95; error dimensionality invariant over forty years), error anisotropy is conserved across paradigm replacements while its direction rotates to the next constraint upstream. A single shared correction vector removes ∼96% (median) of foundation-model elastic-constant error. **Conclusion:** Agreement among models sharing a constraint measures the constraint, not the truth. Error-vector geometry converts benchmarking from scoring into diagnosis, supplies a zero-cost trust rule for ensemble uncertainty, and enables family-level calibration without retraining.

Keywords: model ensembles; error geometry; uncertainty quantification; interatomic potentials; foundation models; density functional theory; verification and validation; benchmarking; formal methods

1 Introduction

Every field that builds models in families uses inter-model agreement as evidence. The practice has long been questioned qualitatively: climate ensembles share construction and history, so their consensus may reflect common structure rather than truth [1, 2], and error correlations between members quantify that dependence [3–5]. In machine learning, independently trained networks

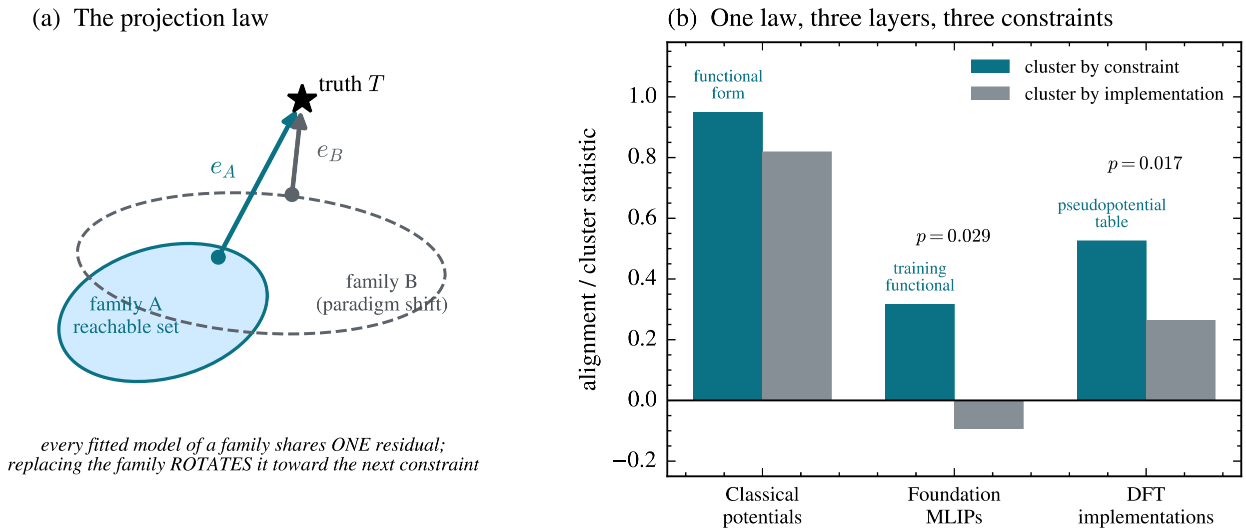

Figure 1: **The projection law and its three-layer test.** (a) A model family's reachable set fixes the residual of every well-fitted member (Theorems 1–2); replacing the family (paradigm shift) rotates the shared residual rather than removing it. (b) Cluster-by-constraint versus cluster-by-implementation statistics at three layers of one epistemic stack. Layer 1 shows within-family vs. pooled reference–prediction correlation (observational); layers 2–3 show the pre-registered cluster statistics with exact permutation p-values; the registered refutation condition (implementation > constraint) was not triggered in either experiment.

agree far beyond what their accuracy predicts [6, 7], ensemble members cluster in prediction space [8], and disagreement tracks the shared bias of the training procedure rather than correctness [9]. In materials simulation specifically — the substrate of this work — ensembles of interatomic potentials have served as error estimators since Frederiksen et al. [10], universal machine-learned interatomic potentials (MLIPs) are known to share a systematic softening inherited from their training data [11, 12], and the reproducibility of density-functional-theory (DFT) implementations is monitored by scalar metrics [13, 14]. What these literatures establish qualitatively, this paper makes geometric, formal, and — in the specific sense of identifying *which* constraint binds — operational.

The central object is the *error vector*: for model i in family F evaluated on observables with reference values, $e_i \in \mathbb{R}^d$ is the vector of relative errors. The law has three parts:

(i) Projection. A model family is a projection operator. Fitting, however performed and by whomever, drives predictions toward the point of the family's reachable set nearest the truth; the residual is a property of the *(family, target)* pair, not of any individual model or modeler. All well-fitted members therefore share one residual direction (Theorems 1–2).

(ii) Gauge. The ensemble's error second moment is a shared bias plus fitting noise; its participation ratio is a closed-form gauge of the systematic fraction (Theorem 3), so low error dimensionality is a necessity for any converged family, at an explicit rate.

(iii) Conservation and rotation. When a modeling paradigm is replaced — when the binding constraint is removed — the error anisotropy does not dissolve into isotropic scatter. It is conserved, and its direction rotates to the next constraint upstream in the epistemic stack (Fig. 1a). This is the part of the law for which we find no antecedent in the climate, forecasting, machine learning, or materials literatures, and it is the part we test hardest (Fig. 1b).

We instantiate the law at three layers of a single epistemic stack — classical interatomic potentials, foundation MLIPs, and the DFT implementations that generate their training data — because this stack offers what few domains can: large open model ensembles, factorial structure (the same architecture trained on two functionals; the same simulation code run with two pseudopotential families), and references at every layer. The empirical discovery paper for the first layer is reported separately [15]; here that layer serves as the observational base of a cross-layer test.

Three methodological commitments distinguish this work, each aligned with the verification, validation, and uncertainty-quantification practices this journal serves. The theory core is *machine-checked*: every theorem below is verified in Lean 4 with zero unproven obligations, and a written contract maps each theorem to the statistic that instantiates it. The two new experiments are *pre-registered*, with thresholds and explicit refutation conditions committed to version control before any model was executed; all threshold misses are reported as failures. And the entire evidence chain is packaged for *tiered replication*: the analysis layer verifies from committed raw data with no machine-learning dependencies in seconds, and the physics layer re-derives the raw data from public checkpoints bit-exactly.

2 Theoretical framework

The mathematics of best approximation is classical [16, 17]; we claim only its epistemic application to model ensembles and the machine-checked instantiation chain. Throughout, E is a real inner-product space (observable space), $s \subseteq E$ a model family's reachable set, $T \in E$ the truth.

Theorem 1 (Normal-cone criterion)**.** *Let s be convex and p a best approximation of T in s (i.e. $p \in s$ and $\|T - p\| \leq \|T - q\|$ for all $q \in s$). Then the residual $T - p$ lies in the normal cone of s at p: $\langle T - p, q - p\rangle \leq 0$ for every $q \in s$.*

Theorem 2 (Consensus)**.** *Best approximations onto a convex family are unique; consequently any two best approximations of the same target share an* identical *residual. The residual — hence any agreement among independently fitted members — is determined by the (family, target) pair alone, and vanishes exactly when the truth lies in the family.*

Agreement among models sharing a constraint is therefore a measurement of the constraint, not evidence of truth: the formal content of the qualitative warnings of Parker [2] and Pirtle et al. [1], and a strengthening of error-correlation dependence measures [3] in that, combined with the factorial designs of §4–5, it identifies *which* constraint binds.

Theorem 3 (Gauge)**.** *Model errors $e = b + \xi$ with shared bias b and isotropic noise of scale σ in d dimensions have error second moment with one eigenvalue $\|b\|^2 + \sigma^2$ (bias axis) and $d-1$ eigenvalues σ^2, and participation ratio*

$$\mathrm{PR}(d, \rho) \;=\; \frac{(\rho + d)^2}{(\rho + 1)^2 + (d - 1)}, \qquad \rho = \|b\|^2/\sigma^2,$$

with $\mathrm{PR}(d, 0) = d$, $1 \leq \mathrm{PR} \leq d$, strict decrease in ρ, and the explicit collapse bound $\mathrm{PR} - 1 \leq 3(d-1)/\rho$ for $\rho \geq d$.

The algebra is a one-line corollary of spiked-covariance theory [18] and the participation ratio is a standard effective-dimension metric [19]; the value here is the consilience it enables (§3) and the machine-checked monotonicity that licenses inverting measured PR into a systematic fraction $\alpha = \rho/(\rho + 1)$.

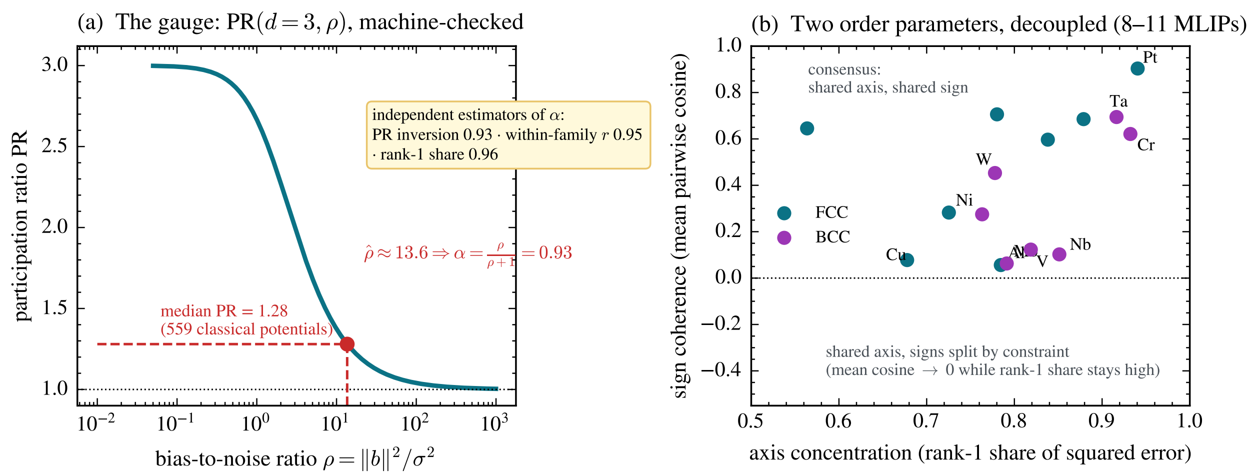

Figure 2: **The gauge and the two order parameters.** (a) PR$(3, \rho)$ (Theorem 3); inverting the classical ensemble's median PR $= 1.28$ gives systematic fraction $\alpha = 0.93$, in agreement with two independent estimators (§3). (b) Per-element foundation-MLIP ensembles ($n = 8$–11 models): axis concentration stays high for every element while sign coherence spans the full range — the decoupling of Theorem 4. Elements whose errors split in sign by training functional (V, Nb, W) sit low-right; consensus elements (Pt, Ta, Au) sit top-right.

Theorem 4 (Ribbon/consensus decoupling)**.** *The participation ratio of a shared-axis ensemble* $\{c_i u\}$ *is* 1 *for every sign pattern of the coefficients* c_i*, while mean pairwise alignment distinguishes them (e.g.* 1 *vs.* $-1/3$ *for three members). PR detects the* axis*; alignment detects* sign coherence*; they are distinct order parameters.*

All four theorems (and the supporting lemmas, ∼29 statements) are machine-checked in Lean 4 against a pinned Mathlib, with zero `sorry` and zero new axioms; the artifact and the theorem-to-statistic contract ship with the replication kit (§10). Figure 2 visualizes the gauge with its classical inversion and the decoupling of the two order parameters in live data.

3 Layer 1: classical interatomic potentials

The observational base [15]: elastic-constant (C_{11}, C_{12}, C_{44}) errors of 559 classical potentials across 15 cubic metals. Three facts matter for the law. First, error dimensionality is low and *stationary*: participation ratios of multi-element potentials occupy $[1.0, 2.1]$ out of 3 with median 1.28, with no trend across forty years of potential development — accuracy improved; geometry did not. You cannot fit your way off a constraint surface. Second, within a functional-form family the errors are nearly identical (within-family reference–prediction correlation $r = 0.95$ across families vs. 0.82 pooled). Third, the gauge locks: the median PR of 1.28 inverts (Theorem 3, d=3) to a systematic fraction $\alpha \approx 0.93$, independently estimated as 0.95 by within-family correlation and 0.96 by rank-one share — three estimators, one parameter, no tuning. The binding constraint at this layer is the functional form itself; the oldest example is exact: central-force pair potentials satisfy the Cauchy relation $C_{12} = C_{44}$ identically, so for any material violating it, every pair potential's error necessarily contains the same forced component — the constraint dictates the direction.

We emphasize the distinction from parameter-space sloppiness: hyper-ribbon geometry was introduced for the parameter manifolds of *single* models [20–23]. The object here is different —

error vectors of an ensemble of *distinct* models in observable space — though the two pictures are related through the projection of parameter manifolds into data space.

4 Layer 2: foundation MLIPs — functional vs. architecture

Foundation MLIPs replace the functional-form constraint with expressive neural networks trained on DFT corpora. The law predicts the anisotropy survives and rotates: the binding constraint becomes the training reference theory. Qualitative inheritance of Perdew–Burke–Ernzerhof (PBE) bias — systematic softening shared by universal MLIPs — has been reported [11, 12]; the contributions here are the directional quantification and the factorial attribution.

Observation. Three PBE-lineage models of unrelated architecture (MACE-MP-0 [24], CHGNet [25], Orb-v3 [26]) commit *the same* elastic-constant errors on FCC metals against experimental references: all-negative, C_{44}-dominant (Au: $-32\%, -27\%, -79\%$), with cross-architecture cosines 0.95–0.99 for Au, Pt, Ta, Ag. An internal control excludes harness artifacts: Ni, through the identical pipeline, has near-zero error. Across 8–11 models per element the rank-one share of squared error is 0.56–0.94 — and where models "disagree" (V, Cr), they disagree by *sign along the same axis* (cosine -0.88), exactly the structure Theorem 4 distinguishes.

Pre-registered factorial test. The MatPES release [27] provides one curated training distribution in two functionals (PBE, r^{2}SCAN) with four architectures trained on each (M3GNet, TensorNet, CHGNet, QET) — a 4×2 crossing of constraint against implementation. We registered thresholds and a refutation condition ("errors cluster by architecture") before executing any model, ran all eight cells through one Born-screened strain-energy harness, and obtained: clustering by functional $S_{\text{func}} = +0.317$ (exact permutation $p = 0.029$ over all 70 labelings) versus clustering by architecture $S_{\text{arch}} = -0.093$ (Fig. 3); the registered rotation prediction confirmed on Au (C_{44} error $-0.33 \to -0.07$) and Pt ($-0.46 \to -0.31$) under r^{2}SCAN; and the dataset-control confirmed (MatPES-PBE models align with MPtrj/OMat-PBE anchors, median cosine 0.66). Two auxiliary thresholds failed as registered (within-functional median 0.654 vs. 0.70; separation 0.085 vs. 0.30): the data revealed *nested* constraints — Cu, Ni, Al errors reorganize by functional (between-functional cosine down to -0.49, the predicted crossover), while the 5d noble metals retain a shared direction across both functionals, bound by correlation physics deeper than the PBE/r^{2}SCAN distinction. The refutation condition was not triggered. The practical reading for transfer learning concurs with Huang et al. [28]: GGA→r^{2}SCAN transfer is hard precisely where the two functionals' error directions genuinely cross over.

5 Layer 3: DFT implementations — pseudopotential vs. code

One layer further up, the functional itself is held fixed. The ACWF verification effort [13, 14] provides equation-of-state parameters (V_0, B_0, B_1) for 384 unary crystals from eleven pseudopotential-based methods and two all-electron codes, all PBE. Against the all-electron average, with the FLEUR–WIEN2k split as the per-system noise floor, the law predicts errors organize by *pseudopotential table* (the approximation actually shared), not by *simulation code* (the implementation). The dataset's factorial structure makes the test sharp: ABINIT appears with three pseudopotential families, Quantum ESPRESSO and CASTEP with two each, while the PseudoDojo-0.4 table is implemented by four independent plane-wave codes.

Pre-registered outcome: same-table–different-code alignment $S_{\text{table}} = +0.526$ versus same-code–different-family $S_{\text{code}} = +0.265$; separation $+0.261$, permutation $p = 0.017$; refutation condition (clustering by code) not triggered (Fig. 4). At pair level, four independent codes sharing

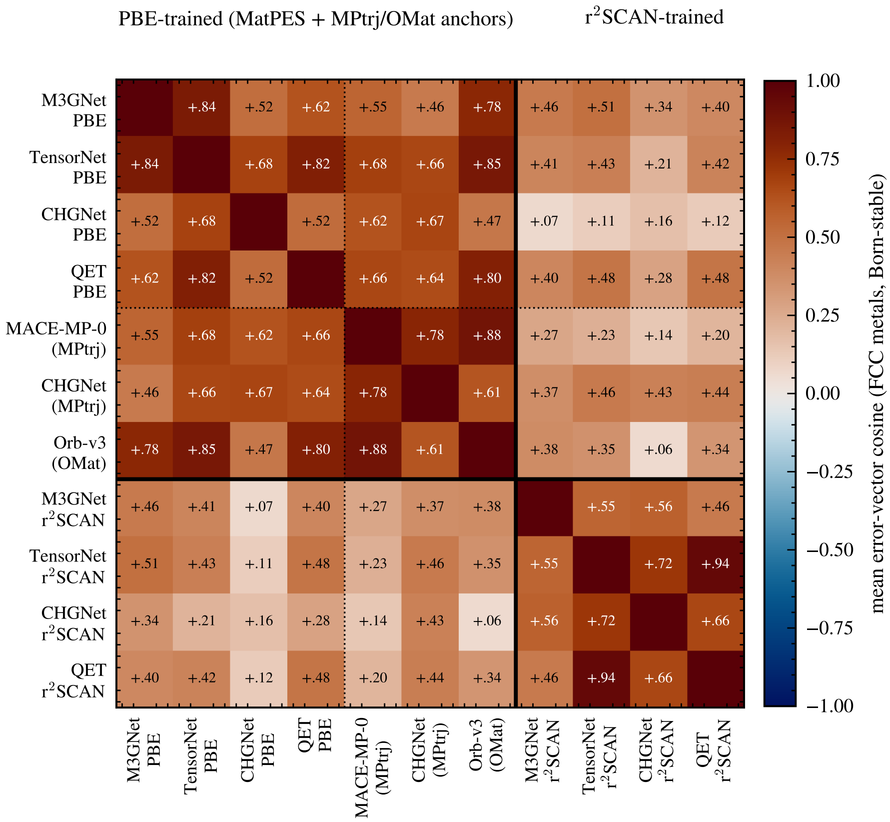

Figure 3: **Foundation-MLIP error geometry organizes by training functional, not architecture.** Mean pairwise error-vector cosine over Born-stable FCC elements for eleven models: four architectures trained on MatPES-PBE, three PBE-lineage anchors (MPtrj/OMat training, dotted separator = dataset axis with functional held fixed), and the same four architectures trained on MatPES-r^{2}SCAN (solid separator). The PBE block coheres across architectures *and* datasets; the cross-functional blocks fade; within-r^{2}SCAN pairs re-cohere (TensorNet$\times$QET +0.94).

PseudoDojo-0.4 align at 0.70–0.87, while ABINIT-with-JTH versus ABINIT-with-PseudoDojo — the same code — drops to 0.21. Two auxiliary predictions failed as registered, both informatively: the within-table consensus median (0.459 vs. 0.50) was dragged down solely by SIESTA, the one localized-orbital-basis code in the group, whose binding constraint is evidently the basis set rather than the shared pseudopotential — the nested-constraint phenomenon again; and the registered regime prediction had the wrong sign because between-family divergence grows on difficult elements, which the pooled statistic conflated. A secondary observation sharpens the law: PAW codes with *different* PAW tables align at only +0.20 — the constraint is the specific frozen-core construction, not the formalism label. Scalar reproducibility metrics [13, 14] cannot see any of this structure; it lives in the directions.

6 The conservation–rotation law

Table 1 assembles the stack. Three layers, three different binding constraints, one geometric law; the two new layers tested under pre-registration with explicit kill conditions, neither triggered. Between layers, the direction *rotates*: classical cross-family alignment has no predictive power for

Table 1: The projection law across one epistemic stack. Each row: an ensemble of models, the constraint the law identifies as binding, the clustering evidence (constraint vs. implementation), and the registered refutation condition's status.

Layer	Ensemble	Binding constraint	Evidence	Kill con
Classical IPs	559 potentials	functional form	within-family r=0.95; PR invariant 40 yr	(observa
Foundation MLIPs	4×2 MatPES + anchors	training functional	S_{func}=+0.317 vs −0.093, p=0.029	not trig
DFT codes	12 ACWF methods	pseudopotential table	S_{table}=+0.526 vs +0.265, p=0.017	not trig

MLIP alignment (Spearman $\rho = 0.26$, $p = 0.34$ across elements), because the constraint moved from functional form to training functional; and the MLIP-layer bias direction is precisely the reference-theory error that layer 3 isolates. Within layers, constraints *nest*: when the registered constraint is removed (r^2SCAN for PBE; plane-wave pseudopotential sharing for SIESTA), residual alignment reveals the next constraint in line (5d correlation physics; basis set). Anisotropy is conserved throughout: at no layer, in no ensemble, under no intervention did errors approach isotropy — participation ratios remained far below ambient dimension in every case, as Theorem 3 requires of any family whose systematic error dominates its fitting scatter.

The closest antecedents are the persistence of shared biases across climate model generations [5, 29], the convergence of training trajectories onto shared low-dimensional manifolds [30], and representation convergence across architectures [31]; none states conservation-plus-rotation as a law, and none possesses the factorial constraint-vs-implementation evidence presented here.

7 Related work

Ensemble dependence across fields. The climate community has the deepest tradition here. Pirtle et al. [1] and Parker [2] argued qualitatively that multi-model agreement may reflect shared construction; Bishop and Abramowitz [3] made dependence operational through pairwise error correlation under the replicate-Earth paradigm, with the elegant null that truly independent models would correlate at 0.5 through the shared observations; [32] review the resulting weighting and subselection machinery, [33] implement performance-plus-independence weighting, [34] traces similarity to literal component replication across modeling centers, and [35, 36] place inter-model error in low-dimensional EOF coordinates. Our normal-cone theorem supplies the mechanism beneath these practices — dependence is not merely shared code but shared *reachable set* — and the factorial designs add what error correlation alone cannot: identification of *which* shared element binds. The persistence of the double-ITCZ bias across three CMIP generations [29] reads, in our terms, as a conservation observation awaiting its rotation experiment.

Machine learning. That independently trained networks agree beyond chance is quantified by error consistency [7] and model similarity [6]; ensemble members cluster in prediction space [8] despite weight-space diversity, which is why deep-ensemble uncertainty [37] inherits the shared component as unaccounted bias — the ambiguity decomposition of Krogh and Vedelsby [38] already showed ensemble spread bounds only the variance term. Disagreement tracks the training procedure's shared bias [9], and accuracy- and agreement-on-the-line [39, 40] exhibit one-dimensional collective structure under distribution shift. Most complementary is underspecification [41]: it studies how equally-fitted models *differ* along directions the family and data leave unconstrained — our noise term ξ — whereas the projection law studies what they *share* — the bias b. Theorem 4 is precisely the statement that these are separate order parameters, measurable separately. Representation convergence across architectures [31] and shared training manifolds [30] are the trajectoryand representation-space cousins of our observable-space result.

Materials simulation. Frederiksen et al. [10] pioneered ensembles of interatomic potentials for error estimation — spread as uncertainty; the law identifies exactly the component spread misses (the shared residual) and supplies the diagnostic for when it dominates (alignment). Shared softening of universal MLIPs is documented [11, 12], and Huang et al. [28] report that GGA→r^2SCAN transfer is hampered by poor inter-functional correlation — our factorial result gives that observation a geometric form. DFT reproducibility is measured by scalar and curve metrics [13, 14]; the directional structure of §5 is invisible to those metrics by construction. Parameter-space sloppiness of single models [20–23] is the intellectual ancestor of this work; we re-emphasize that the object here — error vectors of an ensemble of distinct models in observable space — is different from, though related to, the parameter manifold of any one of them. The participation ratio is a standard effective dimension [19] and the gauge algebra is spiked-covariance theory [18]; the best-approximation mathematics is classical [16, 17].

What is new here is the combination: the error-vector geometry of model ensembles as the measured object; factorial constraint-vs-implementation designs with pre-registered refutation conditions at two layers of one stack; the conservation–rotation law; and a machine-checked theory chain bound to the measurements by a written contract.

8 Discussion: implications for benchmarking, VVUQ, and calibration

Calibration without retraining. Because the shared error is one direction, one correction vector per (element, observable) repairs every model in the family at once: a rank-one correction removes ∼96% (median) of foundation-MLIP elastic error. The correction is family-level: it transfers to models not yet trained, so long as they share the constraint.

A trust rule for ensemble UQ. Cross-model alignment is a zero-cost diagnostic: high alignment ⇒ the ensemble is constraint-bound, calibratable, and its spread *understates* uncertainty; low alignment at the noise floor ⇒ genuine convergence (the Ni control); low alignment far above the floor ⇒ the constraint is heterogeneous (magnetic BCC metals; difficult elements across pseudopotential tables) and no shared correction exists.

Benchmark design. Leaderboards that pool across families average away the very direction that identifies what to fix. Reporting error *directions* stratified by candidate constraints — the factorial pattern used here — converts benchmarking from scoring into diagnosis. This is directly actionable for materials data infrastructure: benchmark records should carry the error vector, not only its norm.

Reading consensus. Where ground truth is delayed (materials discovery campaigns, extrapolative simulation), inter-model agreement should be priced as a measurement of shared constraint; the factorial designs show how to estimate *which* constraint, today, from models alone.

9 Limitations

The formal chain covers convex reachable sets and deterministic bias-plus-noise spectra; smooth non-convex families (local normal cones) and the finite-sample step from noisy ensembles to the exact second moment are standard but unformalized. The MLIP layer uses one validated harness, fifteen cubic elemental metals, and elastic observables; the DFT layer inherits ACWF's protocol choices. Four of nine registered auxiliary thresholds across the two experiments failed and are reported as failures; in each case post-hoc analysis attributes the miss to nested constraints, and those attributions are hypotheses for the next registration round, not confirmed claims. The law itself makes its next predictions cheap to test: an r^{2}SCAN-trained model from a fourth architecture must join the r^{2}SCAN cluster; a plane-wave implementation of any new pseudopotential table must align with its table, not its code; and axis-based statistics (mandated by Theorem 4) should replace signed-cosine thresholds.

10 Reproducibility

Both experiments were pre-registered with thresholds and refutation conditions committed to version control before execution (commits `dffbe595` and `ebf39e33`); all analysis is replayable from a two-tier kit in which Tier 1 verifies every statistic in this paper from committed raw data using only NumPy/SciPy in seconds, and Tier 2 re-derives the raw elastic constants from public checkpoints through a frozen harness (bit-exact on revalidation). The Lean 4 artifact (four theory files, ∼29 machine-checked statements, zero `sorry`, zero new axioms, pinned Mathlib) and the theorem-to-statistic contract accompany the kit. The harness was validated by independent energy- vs. stress-method agreement ($\leq 2.4\%$), bit-exact regression, and strain-window stability. ACWF data are from the public archive of [14]; MatPES checkpoints from [27].

Declarations

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Competing interests. The author declares no competing interests.

Data and code availability. The two-tier replication kit (raw data, analysis scripts, frozen harness), the pre-registration documents with their commit hashes, the Lean 4 formal artifact, and figure-generation scripts are available in the project repository and will be archived with a DOI at acceptance. Third-party data: ACWF verification archive [14]; MatPES checkpoints [27]; OpenKIM/NIST classical-potential data as described in [15].

Author contributions. A.W. conceived the study, performed the research, and wrote the manuscript.

Use of AI assistance. AI assistance (Anthropic Claude) was used under the author's direction for statistical verification and recomputation, Lean proof engineering, literature triage, figure generation, and drafting; all scientific claims and decisions were reviewed and approved by the author, who takes sole responsibility for the content.

References

[1] Zachary Pirtle, Ryan Meyer, and Andrew Hamilton. What does it mean when climate models agree? A case for assessing independence among general circulation models. *Environmental Science & Policy*, 13:351–361, 2010.

[2] Wendy S. Parker. When climate models agree: The significance of robust model predictions. *Philosophy of Science*, 78:579–600, 2011.

[3] Craig H. Bishop and Gab Abramowitz. Climate model dependence and the replicate Earth paradigm. *Climate Dynamics*, 41:885–900, 2013.

[4] James D. Annan and Julia C. Hargreaves. Understanding the CMIP3 multimodel ensemble. *Journal of Climate*, 24:4529–4538, 2011.

[5] Reto Knutti, David Masson, and Andrew Gettelman. Climate model genealogy: Generation CMIP5 and how we got there. *Geophysical Research Letters*, 40:1194–1199, 2013.

[6] Horia Mania, John Miller, Ludwig Schmidt, Moritz Hardt, and Benjamin Recht. Model similarity mitigates test set overuse. In *Advances in Neural Information Processing Systems 32*, 2019.

[7] Robert Geirhos, Kristof Meding, and Felix A. Wichmann. Beyond accuracy: quantifying trial-by-trial behaviour of CNNs and humans by measuring error consistency. In *Advances in Neural Information Processing Systems 33*, 2020.

[8] Stanislav Fort, Huiyi Hu, and Balaji Lakshminarayanan. Deep ensembles: A loss landscape perspective, 2019.

[9] Yiding Jiang, Vaishnavh Nagarajan, Christina Baek, and J. Zico Kolter. Assessing generalization of SGD via disagreement. In *International Conference on Learning Representations*, 2022.

[10] Søren L. Frederiksen, Karsten W. Jacobsen, Kevin S. Brown, and James P. Sethna. Bayesian ensemble approach to error estimation of interatomic potentials. *Physical Review Letters*, 93: 165501, 2004.

[11] Bowen Deng, Yunyeong Choi, Peichen Zhong, Janosh Riebesell, Shashwat Anand, Zhuohan Li, KyuJung Jun, Kristin A. Persson, and Gerbrand Ceder. Systematic softening in universal machine learning interatomic potentials. *npj Computational Materials*, 11:9, 2025.

[12] Haochen Yu, Matteo Giantomassi, Giuliana Materzanini, Junjie Wang, and Gian-Marco Rignanese. Systematic assessment of various universal machine-learning interatomic potentials. *Materials Genome Engineering Advances*, 2:e58, 2024.

[13] Kurt Lejaeghere et al. Reproducibility in density functional theory calculations of solids. *Science*, 352:aad3000, 2016.

[14] Emanuele Bosoni et al. How to verify the precision of density-functional-theory implementations via reproducible and universal workflows. *Nature Reviews Physics*, 6:45–58, 2024.

[15] Alex Welcing. The causal geometry of prediction errors in interatomic potentials: a hyper-ribbon manifold analysis with ecological fallacy detection. Submitted to Integrating Materials and Manufacturing Innovation, 2026.

[16] Frank Deutsch. *Best Approximation in Inner Product Spaces*. Springer, New York, 2001.

[17] David Kinderlehrer and Guido Stampacchia. *An Introduction to Variational Inequalities and Their Applications*. Academic Press, New York, 1980.

[18] Iain M. Johnstone. On the distribution of the largest eigenvalue in principal components analysis. *Annals of Statistics*, 29:295–327, 2001.

[19] Peiran Gao and Surya Ganguli. On simplicity and complexity in the brave new world of large-scale neuroscience. *Current Opinion in Neurobiology*, 32:148–155, 2015.

[20] Mark K. Transtrum, Benjamin B. Machta, and James P. Sethna. Geometry of nonlinear least squares with applications to sloppy models and optimization. *Physical Review E*, 83:036701, 2011.

[21] Benjamin B. Machta, Ricky Chachra, Mark K. Transtrum, and James P. Sethna. Parameter space compression underlies emergent theories and predictive models. *Science*, 342:604–607, 2013.

[22] Katherine N. Quinn, Michael C. Abbott, Mark K. Transtrum, Benjamin B. Machta, and James P. Sethna. Information geometry for multiparameter models: new perspectives on the origin of simplicity. *Reports on Progress in Physics*, 86:035901, 2023.

[23] Yonatan Kurniawan, Cody L. Petrie, Kinamo J. Williams, Mark K. Transtrum, Ellad B. Tadmor, Ryan S. Elliott, Daniel S. Karls, and Mingjian Wen. Bayesian, frequentist, and information geometric approaches to parametric uncertainty quantification of classical empirical interatomic potentials. *Journal of Chemical Physics*, 156:214103, 2022.

[24] Ilyes Batatia et al. A foundation model for atomistic materials chemistry, 2024.

[25] Bowen Deng, Peichen Zhong, KyuJung Jun, Janosh Riebesell, Kevin Han, Christopher J. Bartel, and Gerbrand Ceder. CHGNet as a pretrained universal neural network potential for charge-informed atomistic modelling. *Nature Machine Intelligence*, 5:1031–1041, 2023.

[26] Benjamin Rhodes, Sander Vandenhaute, Vaidotas Šimkus, James Gin, Jonathan Godwin, Tim Duignan, and Mark Neumann. Orb-v3: atomistic simulation at scale, 2025.

[27] Aaron D. Kaplan et al. A foundational potential energy surface dataset for materials, 2025.

[28] Xu Huang, Bowen Deng, Peichen Zhong, Aaron D. Kaplan, Kristin A. Persson, and Gerbrand Ceder. Cross-functional transferability in universal machine learning interatomic potentials, 2025.

[29] Baijun Tian and Xinyu Dong. The double-ITCZ bias in CMIP3, CMIP5, and CMIP6 models based on annual mean precipitation. *Geophysical Research Letters*, 47:e2020GL087232, 2020.

[30] Jialin Mao, Itay Griniasty, Han Kheng Teoh, Rahul Ramesh, Rubing Yang, Mark K. Transtrum, James P. Sethna, and Pratik Chaudhari. The training process of many deep networks explores the same low-dimensional manifold. *Proceedings of the National Academy of Sciences*, 121: e2310002121, 2024.

[31] Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. The platonic representation hypothesis. In *Proceedings of the 41st International Conference on Machine Learning*, 2024.

[32] Gab Abramowitz, Nadja Herger, Ethan Gutmann, et al. ESD reviews: Model dependence in multi-model climate ensembles: weighting, sub-selection and out-of-sample testing. *Earth System Dynamics*, 10:91–105, 2019.

[33] Reto Knutti, Jan Sedláček, Benjamin M. Sanderson, Ruth Lorenz, Erich M. Fischer, and Veronika Eyring. A climate model projection weighting scheme accounting for performance and interdependence. *Geophysical Research Letters*, 44:1909–1918, 2017.

[34] Julien Boé. Interdependency in multimodel climate projections: Component replication and result similarity. *Geophysical Research Letters*, 45:2771–2779, 2018. doi: 10.1002/2017GL076829.

[35] Benjamin M. Sanderson, Reto Knutti, and Peter Caldwell. A representative democracy to reduce interdependency in a multimodel ensemble. *Journal of Climate*, 28:5171–5194, 2015.

[36] Swen Brands. Common error patterns in the regional atmospheric circulation simulated by the CMIP multi-model ensemble. *Geophysical Research Letters*, 49:e2022GL101446, 2022.

[37] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems 30*, 2017.

[38] Anders Krogh and Jesper Vedelsby. Neural network ensembles, cross validation, and active learning. In *Advances in Neural Information Processing Systems 7*. MIT Press, 1995.

[39] John P. Miller, Rohan Taori, Aditi Raghunathan, Shiori Sagawa, Pang Wei Koh, Vaishaal Shankar, Percy Liang, Yair Carmon, and Ludwig Schmidt. Accuracy on the line: On the strong correlation between out-of-distribution and in-distribution generalization. In *Proceedings of the 38th International Conference on Machine Learning*, 2021.

[40] Christina Baek, Yiding Jiang, Aditi Raghunathan, and J. Zico Kolter. Agreement-on-the-line: Predicting the performance of neural networks under distribution shift. In *Advances in Neural Information Processing Systems 35*, 2022.

[41] Alexander D'Amour, Katherine Heller, Dan Moldovan, et al. Underspecification presents challenges for credibility in modern machine learning. *Journal of Machine Learning Research*, 23 (226):1–61, 2022.

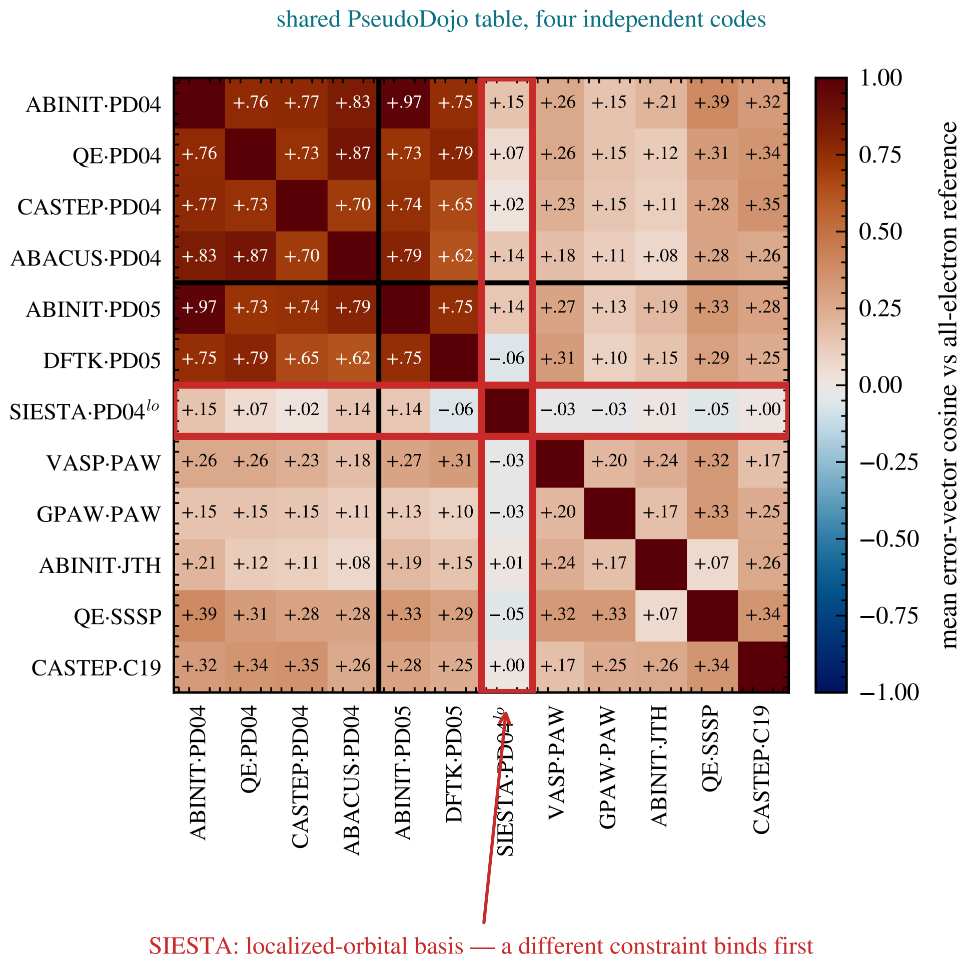

Figure 4: **DFT-implementation errors organize by pseudopotential table, not simulation code.** Mean error-vector cosine in (V_0, B_0, B_1) space against the all-electron reference, over systems above the FLEUR–WIEN2k noise floor (ACWF data, 384 unary crystals). Four independent plane-wave codes sharing the PseudoDojo table align at 0.70–0.87 (dark block); the same code under a different pseudopotential family drops to 0.19–0.35 (e.g. ABINIT-JTH vs. ABINIT-PD04). SIESTA (red frame), the one localized-orbital-basis method, dis-aligns from its own pseudopotential group: a different constraint binds first — the nested-constraint phenomenon.